

80386DX Architecture & Functional Units — MPCA Module 5

Definition: The **Intel 80386DX** is a true **32-bit microprocessor** with a 32-bit internal architecture, a 32-bit Data Bus, and a 32-bit Address Bus, capable of addressing **4 GB of physical memory** and up to **64 Terabytes (TB) of virtual memory**.

1. Detailed Technical Explanation

1. Functional Block Diagram of 80386DX

The 80386DX architecture is organized into **three major sections** containing six distinct functional units:

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80386DX ARCHITECTURE
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| 1. BUS INTERFACE UNIT (BIU): |
|   - 32-bit Address Bus Drivers (A20 - A31, /BE0 - /BE3) |
|   - 32-bit Data Bus Buffer (D0 - D31) |
|   - Bus Control Logic (READY, ADS, /LOCK, BS16) |
| | |
| 2. CENTRAL PROCESSING UNIT (CPU): |
|   - [ Instruction Unit ]: 16-byte Code Prefetch Queue, Instruction |
|                           Decoder (translates opcodes to microcode) |
|   - [ Execution Unit (EU) ]: 32-bit ALU, 8 General Purpose Registers, |
|                               Barrel Shifter (1-cycle 64-bit shift), |
|                               Multiply/Divide Hardware |
| | |
| 3. MEMORY MANAGEMENT UNIT (MMU): |
|   - [ Segmentation Unit ]: Translates Logical Address (Selector:Offset) |
|                           into 32-bit Linear Address via GDT/LDT |
|   - [ Paging Unit ]: Translates 32-bit Linear Address into 32-bit |
|                       Physical Address using 2-level Page Directory |
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2. Address Spaces in 80386DX

- Physical Address Space:** $2^{32} = 4 \text{ GB}$ of physical RAM directly addressable.
- Virtual Memory Space:** $2^{14} \text{ segments} \times 2^{32} \{ \text{bytes/segment} \} = 2^{46} = 64 \{ \text{Terabytes (TB)} \}$ per task.
- Paging:** Supports fixed 4 KB page sizes for virtual memory swapping.

3. Core Concepts & Memory Keywords

- 32-bit Data & Address Bus:** Transfers 32-bit dwords in a single 2-clock bus cycle.
- On-chip MMU:** Integrated hardware Segmentation Unit and Paging Unit on a single silicon die.

- **Barrel Shifter:** Hardware unit performing multi-bit arithmetic/logical shifts in a single clock cycle.
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4. Must-Write Points for Exams

- The 80386DX has a 32-bit data bus and 32-bit address bus (unlike 80386SX which has a 16-bit data bus and 24-bit address bus).
 - Hardware Paging can be dynamically enabled or disabled using the PG bit in Control Register 0 (CR0).
 - The prefetch queue is 16 bytes deep, ensuring continuous pipeline instruction supply.
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5. Quick Recall Flow

80386DX (32-bit) -> 3 Sections (BIU, CPU, MMU) -> Segmentation (Logical to Linear) -> Paging (Linear to Physical) -> 4GB RAM / 64TB Virtual